Data Analytics Hub

Complete quick-reference guide

Part 1	Formulas by category
Part 2	Common Excel errors
Part 3	Keyboard shortcuts (Windows)
Part 4	Handy tips

Note: Shortcuts are for Windows. On Mac, use Cmd instead of Ctrl and Option instead of Alt for most of them. Functions marked 365/2021 need a recent Excel version.

PART 1: FORMULAS

Math & Basic

Function	Syntax / Example	What it does
SUM	`=SUM(A1:A10)`	Adds numbers in a range
AVERAGE	`=AVERAGE(A1:A10)`	Mean of numbers
MIN / MAX	`=MAX(A1:A10)`	Smallest / largest value
COUNT	`=COUNT(A1:A10)`	Counts cells containing numbers
COUNTA	`=COUNTA(A1:A10)`	Counts non-empty cells
COUNTBLANK	`=COUNTBLANK(A1:A10)`	Counts empty cells
ROUND	`=ROUND(A1,2)`	Rounds to given digits
ROUNDUP / ROUNDDOWN	`=ROUNDUP(A1,0)`	Rounds away from / toward zero
CEILING / FLOOR	`=CEILING(A1,5)`	Rounds up / down to a multiple
INT	`=INT(A1)`	Rounds down to nearest integer
TRUNC	`=TRUNC(A1,1)`	Cuts off decimals without rounding
MOD	`=MOD(10,3)`	Remainder after division
ABS	`=ABS(A1)`	Absolute value
POWER	`=POWER(2,3)`	Number raised to a power
SQRT	`=SQRT(A1)`	Square root
PRODUCT	`=PRODUCT(A1:A5)`	Multiplies numbers
SUMPRODUCT	`=SUMPRODUCT(A1:A5,B1:B5)`	Multiplies arrays then sums
RAND / RANDBETWEEN	`=RANDBETWEEN(1,100)`	Random number
LARGE / SMALL	`=LARGE(A1:A10,2)`	K-th largest / smallest value
MEDIAN / MODE	`=MEDIAN(A1:A10)`	Middle value / most frequent value
SUBTOTAL	`=SUBTOTAL(9,A1:A10)`	Sum/avg etc. that ignores filtered rows
AGGREGATE	`=AGGREGATE(9,6,A1:A10)`	Like SUBTOTAL, can ignore errors
RANK	`=RANK(A1,A$1:A$10)`	Rank of a number in a list
PERCENTILE	`=PERCENTILE(A1:A10,0.9)`	K-th percentile
STDEV.S / VAR.S	`=STDEV.S(A1:A10)`	Sample standard deviation / variance

Logical

Function	Syntax / Example	What it does
IF	`=IF(A1>50,"Pass","Fail")`	Returns value based on a condition
IFS	`=IFS(A1>90,"A",A1>80,"B",TRUE,"C")`	Multiple conditions, no nesting
AND	`=AND(A1>0,B1>0)`	TRUE if all conditions true
OR	`=OR(A1>0,B1>0)`	TRUE if any condition true
NOT	`=NOT(A1>5)`	Reverses TRUE/FALSE
XOR	`=XOR(A1>0,B1>0)`	TRUE if odd number of conditions true
IFERROR	`=IFERROR(A1/B1,0)`	Returns fallback if error
IFNA	`=IFNA(VLOOKUP(...),"NA")`	Fallback only for #N/A
SWITCH	`=SWITCH(A1,1,"Mon",2,"Tue","Other")`	Matches a value to a list of results

Function	Syntax / Example	What it does
TRUE / FALSE	`=TRUE()`	Boolean constants

Conditional Sum / Count

Function	Syntax / Example	What it does
SUMIF	`=SUMIF(A:A,"East",B:B)`	Sum with one condition
SUMIFS	`=SUMIFS(C:C,A:A,"East",B:B,">100")`	Sum with multiple conditions
COUNTIF	`=COUNTIF(A:A,">50")`	Count with one condition
COUNTIFS	`=COUNTIFS(A:A,"East",B:B,">100")`	Count with multiple conditions
AVERAGEIF	`=AVERAGEIF(A:A,"East",B:B)`	Average with one condition
AVERAGEIFS	`=AVERAGEIFS(C:C,A:A,"East",B:B,">100")`	Average with multiple conditions
MAXIFS / MINIFS	`=MAXIFS(C:C,A:A,"East")`	Max / min with conditions

Lookup & Reference

Function	Syntax / Example	What it does
VLOOKUP	`=VLOOKUP(A1,D:F,3,FALSE)`	Vertical lookup, exact match
HLOOKUP	`=HLOOKUP(A1,A5:F8,2,FALSE)`	Horizontal lookup
XLOOKUP	`=XLOOKUP(A1,D:D,F:F,"Not found")`	Modern lookup, any direction
XMATCH	`=XMATCH(A1,D:D)`	Modern MATCH
INDEX	`=INDEX(A1:C10,3,2)`	Value at row/column of a range
MATCH	`=MATCH(A1,D1:D10,0)`	Position of a value in a range
INDEX+MATCH	`=INDEX(F:F,MATCH(A1,D:D,0))`	Flexible lookup combo
LOOKUP	`=LOOKUP(A1,D:D,F:F)`	Simple sorted lookup
CHOOSE	`=CHOOSE(2,"A","B","C")`	Picks value by index number
OFFSET	`=OFFSET(A1,2,1,3,1)`	Range shifted from a reference
INDIRECT	`=INDIRECT("Sheet2!A1")`	Reference from text
ROW / COLUMN	`=ROW(A5)`	Row / column number
ROWS / COLUMNS	`=ROWS(A1:A10)`	Number of rows / columns
ADDRESS	`=ADDRESS(2,3)`	Cell address as text
TRANSPOSE	`=TRANSPOSE(A1:C3)`	Swaps rows and columns
HYPERLINK	`=HYPERLINK("https://x.com","Go")`	Creates a clickable link

Text

Function	Syntax / Example	What it does
LEFT / RIGHT	`=LEFT(A1,3)`	First / last N characters
MID	`=MID(A1,2,4)`	Characters from the middle
LEN	`=LEN(A1)`	Number of characters
FIND	`=FIND("a",A1)`	Position of text (case-sensitive)
SEARCH	`=SEARCH("a",A1)`	Position of text (not case-sensitive)
SUBSTITUTE	`=SUBSTITUTE(A1,"old","new")`	Replaces specific text
REPLACE	`=REPLACE(A1,1,3,"XYZ")`	Replaces by position
TRIM	`=TRIM(A1)`	Removes extra spaces

Function	Syntax / Example	What it does
CLEAN	`=CLEAN(A1)`	Removes non-printing characters
UPPER / LOWER / PROPER	`=PROPER(A1)`	Changes text case
CONCATENATE / CONCAT	`=CONCAT(A1,B1)`	Joins text
TEXTJOIN	`=TEXTJOIN(", ",TRUE,A1:A5)`	Joins with delimiter, skips blanks
TEXT	`=TEXT(A1,"dd-mmm-yyyy")`	Formats number/date as text
VALUE	`=VALUE(A1)`	Text to number
REPT	`=REPT("*",5)`	Repeats text
EXACT	`=EXACT(A1,B1)`	Case-sensitive comparison
TEXTSPLIT	`=TEXTSPLIT(A1,",")`	Splits text into cells (365)
TEXTBEFORE / TEXTAFTER	`=TEXTBEFORE(A1,"@")`	Text before / after a delimiter (365)
CHAR / CODE	`=CHAR(65)`	Character from code / code from character
NUMBERVALUE	`=NUMBERVALUE("1.234,5",",",".")`	Text to number with locale

Date & Time

Function	Syntax / Example	What it does
TODAY	`=TODAY()`	Current date
NOW	`=NOW()`	Current date and time
DATE	`=DATE(2026,9,30)`	Builds a date
TIME	`=TIME(14,30,0)`	Builds a time
YEAR / MONTH / DAY	`=YEAR(A1)`	Extracts part of a date
HOUR / MINUTE / SECOND	`=HOUR(A1)`	Extracts part of a time
WEEKDAY	`=WEEKDAY(A1,2)`	Day number of the week
WEEKNUM	`=WEEKNUM(A1)`	Week number of the year
EDATE	`=EDATE(A1,3)`	Date N months later/earlier
EOMONTH	`=EOMONTH(A1,0)`	Last day of the month
DATEDIF	`=DATEDIF(A1,B1,"Y")`	Difference in Y / M / D
DAYS	`=DAYS(B1,A1)`	Days between two dates
NETWORKDAYS	`=NETWORKDAYS(A1,B1)`	Working days between dates
WORKDAY	`=WORKDAY(A1,10)`	Date after N working days
YEARFRAC	`=YEARFRAC(A1,B1)`	Fraction of year between dates
DATEVALUE / TIMEVALUE	`=DATEVALUE("30-09-2026")`	Text to date / time

Financial

Function	Syntax / Example	What it does
PMT	`=PMT(8%/12,60,-500000)`	Loan payment per period
IPMT / PPMT	`=IPMT(8%/12,1,60,-500000)`	Interest / principal part of a payment
FV	`=FV(7%/12,120,-5000)`	Future value
PV	`=PV(7%/12,120,-5000)`	Present value
NPV	`=NPV(10%,B1:B5)`	Net present value

Function	Syntax / Example	What it does
IRR	`=IRR(B1:B6)`	Internal rate of return
XNPV / XIRR	`=XIRR(B1:B6,A1:A6)`	NPV / IRR with dates
RATE	`=RATE(60,-10000,500000)`	Interest rate per period
NPER	`=NPER(8%/12,-10000,500000)`	Number of periods
SLN / DB / DDB	`=SLN(10000,1000,5)`	Depreciation methods
EFFECT / NOMINAL	`=EFFECT(12%,12)`	Effective / nominal interest rate

Dynamic Arrays (365 / 2021)

Function	Syntax / Example	What it does
FILTER	`=FILTER(A2:C20,B2:B20>100)`	Returns rows meeting a condition
SORT	`=SORT(A2:C20,2,-1)`	Sorts a range
SORTBY	`=SORTBY(A2:A20,B2:B20,-1)`	Sorts by another range
UNIQUE	`=UNIQUE(A2:A20)`	Unique values
SEQUENCE	`=SEQUENCE(5,1,1,1)`	Generates number series
RANDARRAY	`=RANDARRAY(3,3,1,100,TRUE)`	Array of random numbers
TAKE / DROP	`=TAKE(A1:C10,3)`	First/last rows or columns / removes them
VSTACK / HSTACK	`=VSTACK(A1:B3,D1:E3)`	Stacks ranges vertically / horizontally
TOCOL / TOROW	`=TOCOL(A1:C3)`	Array to single column / row
LET	`=LET(x,A1*2,x+1)`	Names values inside a formula
LAMBDA	`=LAMBDA(x,x*2)(5)`	Custom reusable function
MAP / REDUCE / SCAN	`=MAP(A1:A5,LAMBDA(x,x*2))`	Apply LAMBDA over arrays
BYROW / BYCOL	`=BYROW(A1:C3,LAMBDA(r,SUM(r)))`	Apply LAMBDA row-wise / column-wise
Spill operator #	`=SUM(A1#)`	References the whole spilled range

Information & Others

Function	Syntax / Example	What it does
ISBLANK	`=ISBLANK(A1)`	TRUE if empty
ISNUMBER / ISTEXT	`=ISNUMBER(A1)`	Checks data type
ISERROR / ISERR / ISNA	`=ISERROR(A1)`	Checks for errors
ISEVEN / ISODD	`=ISEVEN(A1)`	Checks even / odd
ISFORMULA	`=ISFORMULA(A1)`	TRUE if cell has a formula
TYPE	`=TYPE(A1)`	Data type code
N	`=N(A1)`	Converts value to number
NA	`=NA()`	Returns #N/A
CELL	`=CELL("filename")`	Info about a cell
FORMULATEXT	`=FORMULATEXT(A1)`	Shows the formula as text
SHEET / SHEETS	`=SHEET()`	Sheet number / count
GETPIVOTDATA	`=GETPIVOTDATA("Sales",A3)`	Pulls data from a PivotTable

PART 2: COMMON EXCEL ERRORS

Error	Meaning
#DIV/0!	Dividing by zero or empty cell
#N/A	Value not found (lookup failed)
#NAME?	Unrecognised function name or text
#NULL!	Incorrect range operator / intersection
#NUM!	Invalid number in a formula
#REF!	Invalid cell reference (deleted cell)
#VALUE!	Wrong type of argument
#SPILL!	Spill range blocked
#####	Column too narrow for the value

PART 3: KEYBOARD SHORTCUTS

General

Shortcut	Action
Ctrl + N	New workbook
Ctrl + O	Open workbook
Ctrl + S	Save
F12	Save As
Ctrl + P	Print
Ctrl + W / Ctrl + F4	Close workbook
Alt + F4	Close Excel
Ctrl + Z	Undo
Ctrl + Y / F4	Redo / repeat last action
Ctrl + C / X / V	Copy / Cut / Paste
Ctrl + Alt + V	Paste Special
Ctrl + F	Find
Ctrl + H	Find and Replace
Ctrl + G / F5	Go To
F1	Help
F7	Spell check
F9	Calculate all workbooks
Shift + F9	Calculate active sheet
Esc	Cancel entry
Ctrl + Shift + L	Toggle filter
Ctrl + T	Create a table
Alt	Show ribbon key tips
Ctrl + F1	Show / hide ribbon
Ctrl + Shift + U	Expand / collapse formula bar

Navigation

Shortcut	Action
Arrow keys	Move one cell
Ctrl + Arrow	Jump to edge of data region
Ctrl + Home	Go to A1
Ctrl + End	Go to last used cell
Home	Start of row
Page Up / Page Down	Move one screen up / down
Alt + Page Up / Down	Move one screen left / right
Ctrl + Page Up / Down	Previous / next sheet
Ctrl + Tab	Next open workbook
Tab / Shift + Tab	Move right / left
Enter / Shift + Enter	Move down / up
Ctrl + Backspace	Show active cell

Shortcut	Action
F6	Switch between ribbon, sheet, and panes

Selection

Shortcut	Action
Ctrl + A	Select all (press twice inside data)
Ctrl + Space	Select entire column
Shift + Space	Select entire row
Shift + Arrow	Extend selection by one cell
Ctrl + Shift + Arrow	Extend selection to end of data
Ctrl + Shift + Home / End	Select to start of sheet / last used cell
Ctrl + Shift + *	Select current region
Shift + F8	Add non-adjacent cells to selection
F8	Extend selection mode
Ctrl + [	Select direct precedents
Ctrl +]	Select direct dependents
**Ctrl + **	Select row differences
Alt + ;	Select visible cells only
Ctrl + Click	Select non-adjacent cells

Editing & Data Entry

Shortcut	Action
F2	Edit active cell
Alt + Enter	New line inside a cell
Ctrl + Enter	Fill selected cells with the entry
Ctrl + D	Fill down
Ctrl + R	Fill right
Ctrl + E	Flash Fill
Ctrl + ;	Insert current date
Ctrl + Shift + :	Insert current time
Ctrl + '	Copy formula from cell above
Ctrl + Shift + "	Copy value from cell above
Ctrl + + (plus)	Insert cells / rows / columns
Ctrl + - (minus)	Delete cells / rows / columns
Delete	Clear cell contents
Ctrl + K	Insert hyperlink
Shift + F2	Insert / edit comment
Ctrl + Shift + F2	Insert / edit threaded comment
Alt + Down Arrow	Open drop-down list / AutoComplete list
Alt + =	AutoSum
Ctrl + 9 / Ctrl + Shift + 9	Hide / unhide rows
Ctrl + 0 / Ctrl + Shift + 0	Hide / unhide columns
Alt + Shift + Right / Left	Group / ungroup rows or columns

Formulas

Shortcut	Action
=	Start a formula
F4	Cycle absolute / relative references (A1)
Ctrl + `	Show / hide formulas
Shift + F3	Insert Function dialog
Ctrl + Shift + A	Insert argument names
Ctrl + A (in formula)	Function Arguments dialog
F3	Paste a named range
Ctrl + F3	Name Manager
Ctrl + Shift + F3	Create names from selection
Ctrl + Shift + Enter	Enter as array formula (older Excel)
F9 (in formula)	Evaluate selected part of formula
Ctrl + [	Jump to precedent cells
Alt + M + P / D	Trace precedents / dependents
Alt + M + A	Remove trace arrows
Alt + M + V	Evaluate formula
Tab (in autocomplete)	Accept suggested function

Formatting

Shortcut	Action
Ctrl + 1	Format Cells dialog
Ctrl + B / I / U	Bold / Italic / Underline
Ctrl + 5	Strikethrough
Ctrl + Shift + ~	General number format
Ctrl + Shift + !	Number format with 2 decimals
Ctrl + Shift + $	Currency format
Ctrl + Shift + %	Percentage format
Ctrl + Shift + ^	Scientific format
Ctrl + Shift + #	Date format
Ctrl + Shift + @	Time format
Ctrl + Shift + &	Apply outline border
Ctrl + Shift + _	Remove borders
Alt + H + H	Fill colour
Alt + H + F + C	Font colour
Alt + H + A + C / L / R	Center / left / right align
Alt + H + M + C	Merge and center
Alt + H + W	Wrap text
Alt + H + O + I	Auto-fit column width
Alt + H + O + A	Auto-fit row height
Ctrl + Shift + F	Font tab of Format Cells
Alt + '	Style dialog

Worksheets, Windows & Data Tools

Shortcut	Action
Shift + F11	Insert new worksheet
Alt + H + I + S	Insert sheet (ribbon)
Alt + H + D + S	Delete sheet
Alt + H + O + R	Rename sheet
Alt + H + O + M	Move / copy sheet
Ctrl + Shift + Page Up / Down	Select multiple sheets
Alt + W + F + F	Freeze panes
Alt + W + S	Split window
Ctrl + F6	Switch to next workbook window
Alt + N + V	Insert PivotTable
Alt + D + P	PivotTable wizard (classic)
Alt + A + S + S	Sort dialog
Alt + A + T	Toggle filter
Alt + A + V + V	Data Validation
Alt + A + M	Remove duplicates
Alt + A + E	Text to Columns
Alt + A + W + G	Goal Seek
Alt + N + C	Insert column chart
F11	Create chart on new sheet
Alt + F1	Create embedded chart from selection
Alt + F8	Macro dialog
Alt + F11	Open VBA editor
Alt + T + O	Excel Options
Ctrl + Q	Quick Analysis

PART 4: HANDY TIPS

- Use $ to lock references: A1 (both), A$1 (row), $A1 (column). Press F4 to cycle.
- Prefer XLOOKUP over VLOOKUP when available; it looks left and right and defaults to exact match.
- Wildcards work in SUMIF/COUNTIF/MATCH: * matches any text, ? matches one character.
- Operators: + - * / ^ (power), & (join text), = < > <= >= <> (comparison).
- Order of operations: brackets, exponents, multiply/divide, add/subtract.
- Use named ranges and tables (Ctrl + T) to make formulas easier to read.
- Press Ctrl + ` to see all formulas on a sheet at once when auditing.